51.2 V · 100 AH · 5.12 KWH · LITHIUM IRON PHOSPHATE

OLB-5.12 Series Residential Storage Battery

A safe, long-life LFP battery pack for homes, small offices and micro off-grid systems. Choose wall-mount or 19-inch rack-mount, and parallel up to 10 packs for 51.2 kWh of storage — a natural partner to the Olitec OLI-1H hybrid inverter.

OLB-5.12W · Wall-mount | OLB-5.12R · Rack-mount (3U)



5.12 kWh per pack, 100 Ah @ 51.2 V

4,000 cycles @ 80 % DoD, 25 °C, to 70 % SoH

10 packs parallel · up to 51.2 kWh

Why OLB-5.12

$LiFePO_4$ safety
Thermally stable, cobalt-free chemistry; cells certified to BIS, UN 38.3, IEC 62619 and UL 1973.

Intelligent BMS
Cell-level voltage, current and temperature monitoring with over-charge, over-discharge, over-current and short-circuit protection.

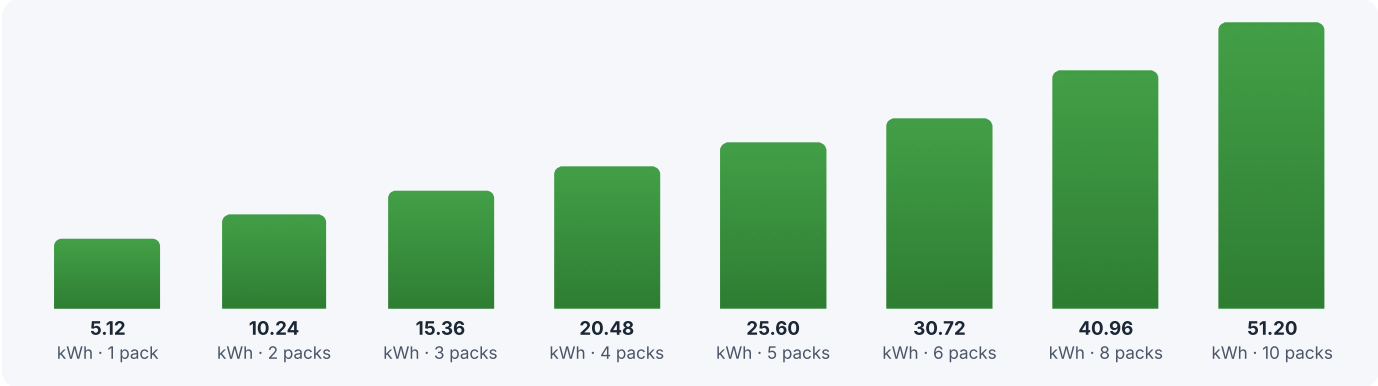
Scales with you
Start with one pack and parallel up to 10 — 5.12 kWh to 51.2 kWh on the same 51.2 V bus.

5 kW continuous
50 A recommended, 100 A maximum continuous charge and discharge per pack; 300 A peak for 10 s.

Open communication
RS485 × 3, CAN × 1 and RS232 × 1 for inverter integration (Olitec OLI-1H plug-and-play) and monitoring.

Flexible installation
Slim 198 mm wall-mount or standard 3U / 19″ rack-mount; passive cooling, no fans to maintain.

Modular capacity



A module power combiner box is recommended for parallel systems. System operating voltage 43.2 – 57.6 V.

Olitec Solar Private Limited · No. 5/63, Gateway House, House of Hiranandani, Egattur, OMR, Chennai – 600 130, Tamil Nadu, India
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Pack specifications

51.2 V LFP

Parameter	OLB-5.12W (Wall-mount)	OLB-5.12R (Rack-mount)
Nominal voltage	51.2 V	51.2 V
Nominal capacity	100 Ah	100 Ah
Rated energy	5.12 kWh	5.12 kWh
Operating voltage range	44.8 – 57.6 V	44.8 – 57.6 V
Recommended charge / discharge current	50 A continuous	50 A continuous
Max. charge / discharge current	100 A	100 A
Charging temperature range	0 °C to 50 °C	0 °C to 50 °C
Discharging temperature range	0 °C to 55 °C	0 °C to 55 °C
Cooling	Passive	Passive
Internal resistance	≤ 0.25 mΩ	≤ 0.25 mΩ
Cycle life	4,000 cycles @ 25 °C, 80 % DoD, 0.5C / 0.5C, to 70 % SoH	4,000 cycles @ 25 °C, 80 % DoD, 0.5C / 0.5C, to 70 % SoH
Dimensions (H × W × D)	644 × 410 × 198 mm	133.5 × 482 × 553 mm (3U, 19″ rack)
Weight	54 ± 2 kg	45 ± 2 kg
Ingress protection	IP20	IP20
Communication	RS485 × 3, CAN × 1, RS232 × 1	RS485 × 3, CAN × 1, RS232 × 1
Storage condition	15 – 35 °C; self-discharge ≈ 7 % / month; recharge every 3 months	15 – 35 °C; self-discharge ≈ 7 % / month; recharge every 3 months
Mounting	Wall-mounted	Rack-mounted

Cell specifications

Parameter	Specification
Cell chemistry	Lithium iron phosphate (LFP / $LiFePO_4$)
Nominal rating	3.2 V, 100 Ah (320 Wh per cell)
AC internal impedance	≤ 0.6 mΩ (0.3 mΩ typical)
Standard continuous charge / discharge current	50 A / 50 A (0.5C / 0.5C)
Max. continuous charge / discharge current	100 A / 100 A (1C / 1C)
Peak charge / discharge current (10 s, non-cyclic)	200 A / 300 A (2C / 3C)
Cell-level certification	BIS, UN 38.3, IEC 62619, UL 1973

Parallel system specifications

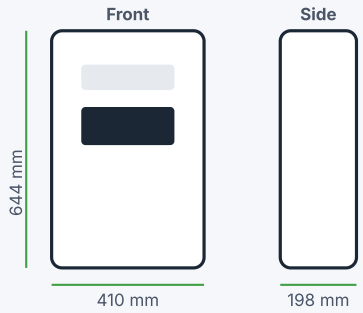
Parameter	Specification
Nominal capacity	100 Ah × N (N = number of packs, max. 10)
Usable energy range	5.12 kWh to 51.2 kWh
Nominal voltage	51.2 V
Recommended charge / discharge current	50 A continuous
Max. charge / discharge current	100 A
Operating voltage	43.2 – 57.6 V
Operating temperature	0 – 50 °C (charging), 0 – 55 °C (discharging)
System weight	Number of packs × unit weight
Dimensions	As per general arrangement drawing / mounting layout
Parallel connection	Module power combiner box recommended for parallel systems

Cycle life tested at 25 °C, 80 % depth of discharge, 0.5C charge / 0.5C discharge to 70 % state of health. Store between 15 – 35 °C; self-discharge approximately 7 % per month — recharge at least every 3 months. All specifications subject to change without prior notice.

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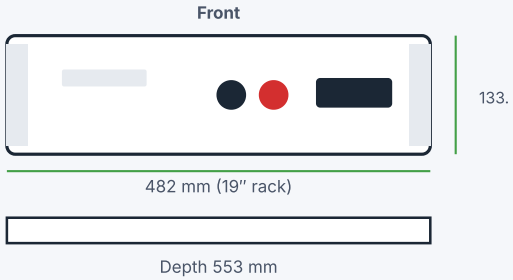
Dimensions

OLB-5.12W · Wall-mount



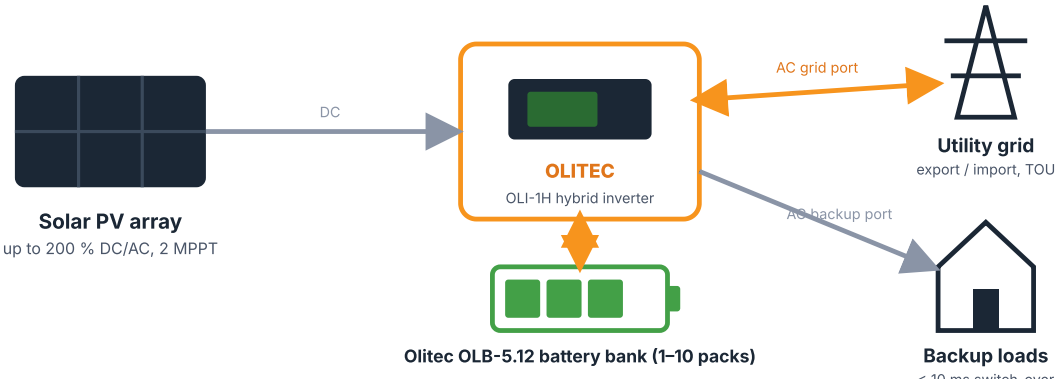
644 × 410 × 198 mm (H × W × D) · 54 ± 2 kg · IP20

OLB-5.12R · Rack-mount



133.5 × 482 × 553 mm (H × W × D) · 45 ± 2 kg · IP20

System overview



Compatible inverters

- **Olitec OLI-1H series** single-phase hybrid, 3 – 6 kW (CAN communication, plug-and-play).
- Any 48 V-class hybrid / off-grid inverter supporting lithium batteries over CAN or RS485.
- Lead-acid-mode operation is possible with voltage-based charging (44.8 – 57.6 V).

Installation notes

- Indoor installation (IP20); ambient 0 – 50 °C charging, 0 – 55 °C discharging.
- Use the module power combiner box when paralleling two or more packs.
- Keep 100 mm clearance around the pack for ventilation; passive cooling only.
- Wall-mount requires a load-bearing wall rated for ≥ 60 kg; rack-mount fits a standard 19″ cabinet (3U).

Certifications

BIS
Cell-level certification under the Indian compulsory registration scheme.

UN 38.3
Transport safety testing for lithium cells.

IEC 62619
Safety requirements for industrial secondary lithium cells.

UL 1973
Batteries for stationary and motive auxiliary power applications.

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